

It's the Era of Conversational AI



Conversational AI helps customers interact with computer applications like chatbots just the way they would with humans. Let's explore this domain and take a look at what the tech giants are offering in this space.

Conversational artificial intelligence (AI) is today being used to implement various new age AI solutions like chatbots, virtual assistants, and contact centres, to name a few. Cloud based architectures like Azure AI, AWS ML or GCP ML provide many services suitable for building a chatbot combined with other native cloud services. AWS has even provided pre-build CloudFormation templates from Marketplace to swiftly develop a serverless chatbot service.

Smart devices like Amazon Echo and Apple HomePod use voice based intelligent virtual assistants (IVAs) to

automate regular tasks. Amazon Echo uses Alexa, while Apple HomePod uses Siri. These simple software agents can take voice commands and execute tasks using an Internet connection and API interfaces.

Technical architecture of conversational AI chatbots

Typically, all IVA interfaces work using natural language processing (NLP) by segmenting audio inputs. For example, a command like "Siri, call Alan on his home number," will be split into each word using automatic speech recognition (ASR). The dictionary of

phonetics will then be searched for a suitable mapped pattern to get the relevant action to execute the command by using an API interface. After executing the action, the response will be formulated using a text-to-speech (TTS) interface.

A chatbot is an IVA but is different from Siri/Alexa devices. A chatbot or conversational assistant is a dialogue based system that takes continuous inputs and uses previous chat messages to contextualise the response. Alexa/Siri are service agents that take commands and have an event driven approach, where a voice command is the event.